

Course Code	EC24P
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Roll Number	23104B0003
Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	5 th September 2025
Date of Submission	12 th September 2025

EXPERIMENT NO.6

Estimated time to complete this experiment: 02 hours.

Objective: To simulate and find the rise time and fall time of CMOS inverter using gpdk180.

CO to be achieved: CO2.

Expected Outcome of Experiment:

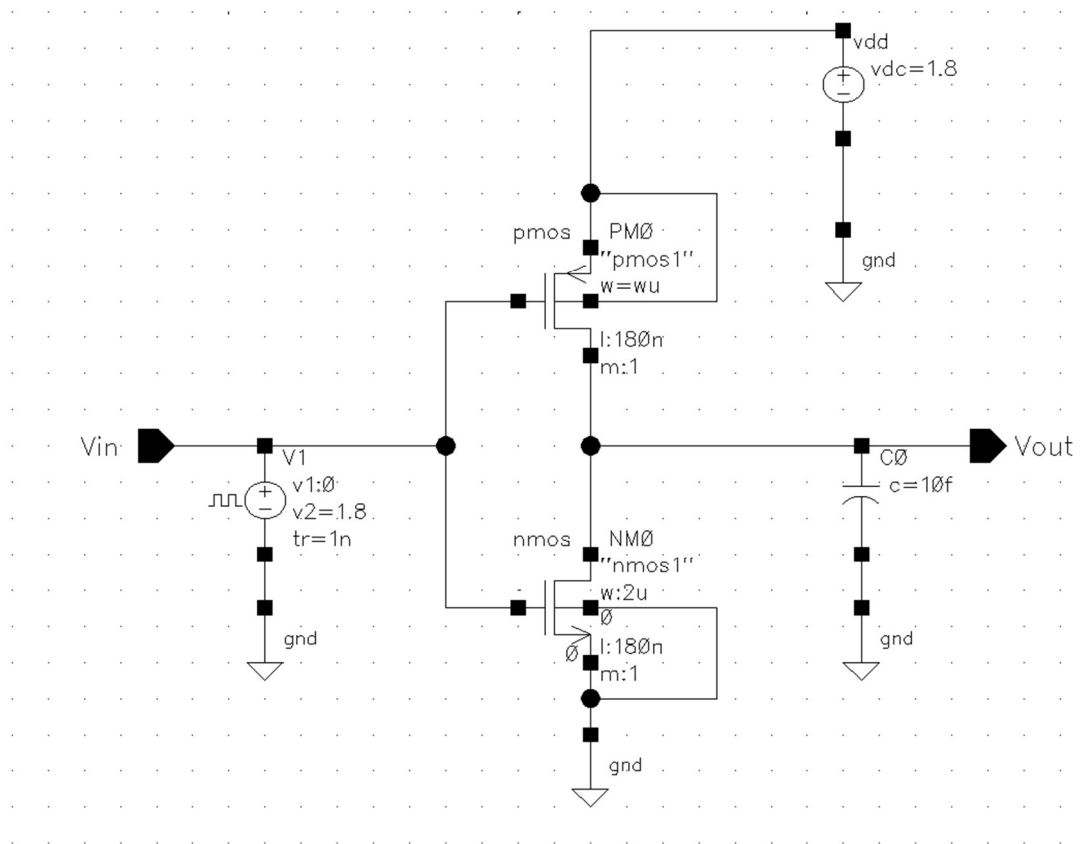
- Students will learn cadence virtuoso.
- Students will be able to find rise time and fall time of asymmetric CMOS inverter.
- Students will be able to find propagation delay time of CMOS Inverter.

Pre Lab/ Prior Concept: Knowledge of working of MOSFET and CMOS inverter transient analysis.

Apparatus/TOOL:

- PC
- CADENCE VIRTUOSO.

Circuit Diagram:



5) Procedure.

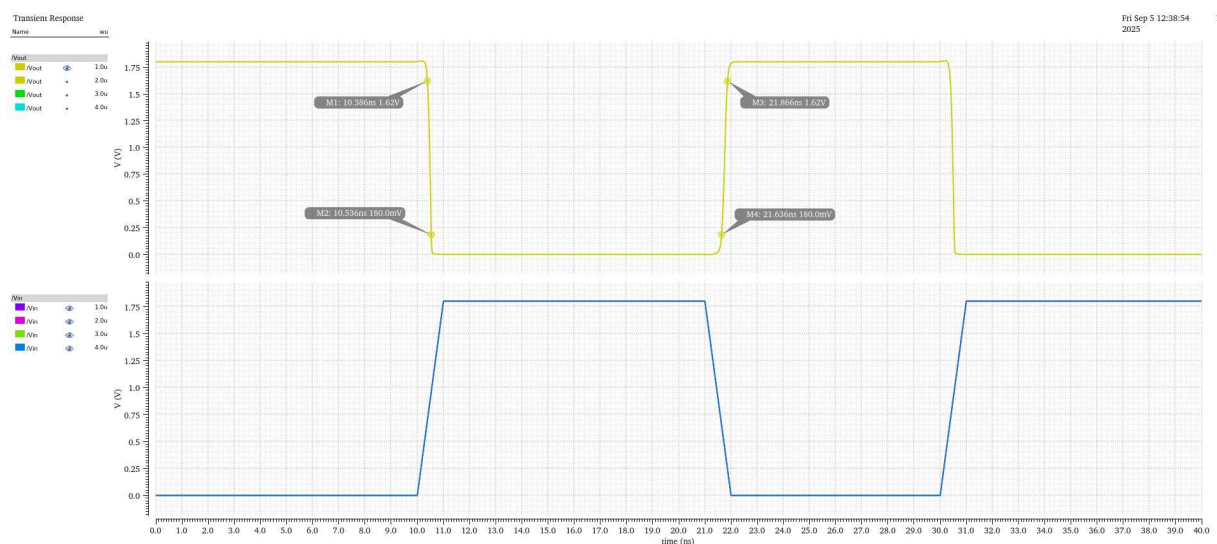
- Open the cadence virtuoso tool in Red-Hat Linux. The commands are
 - csh
 - source /home/install/cshrc
 - virtuoso.
- Create the Library and cell view
- Open log window using terminal.
- Click on the new option, now create library with gpdk180 from technology library.

- Again click the on new option, and set the cell view.
- Cadence Virtuoso dashboard will be opened.
- For characterization of 180nm CMOS Inverter.
- Select NMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width $W=2\text{ }\mu\text{m}$ (micro-meter).
- Select PMOS transistor from add instance and select its channel length as 180nm (Nano meter) and channel width as $W= 2\text{ }\mu\text{m}$ (micro-meter).
- Construct the circuit shown in circuit diagram in Cadence.
- Take $VDD=1.8\text{V}$.
- Apply input as V_{PULSE} with following specification.
 - Voltage 1= 0V
 - Voltage 2= 1.8V
 - Delay time= 10ns
 - Rise time = 1ns
 - Fall time = 1ns
 - Pulse width = 10ns .
 - Period = 20ns .
- Then perform the Transient analysis and plot the input and output waveform.
- Using input and output waveform calculate the following parameters.
 - T_{RISE}
 - T_{FALL}

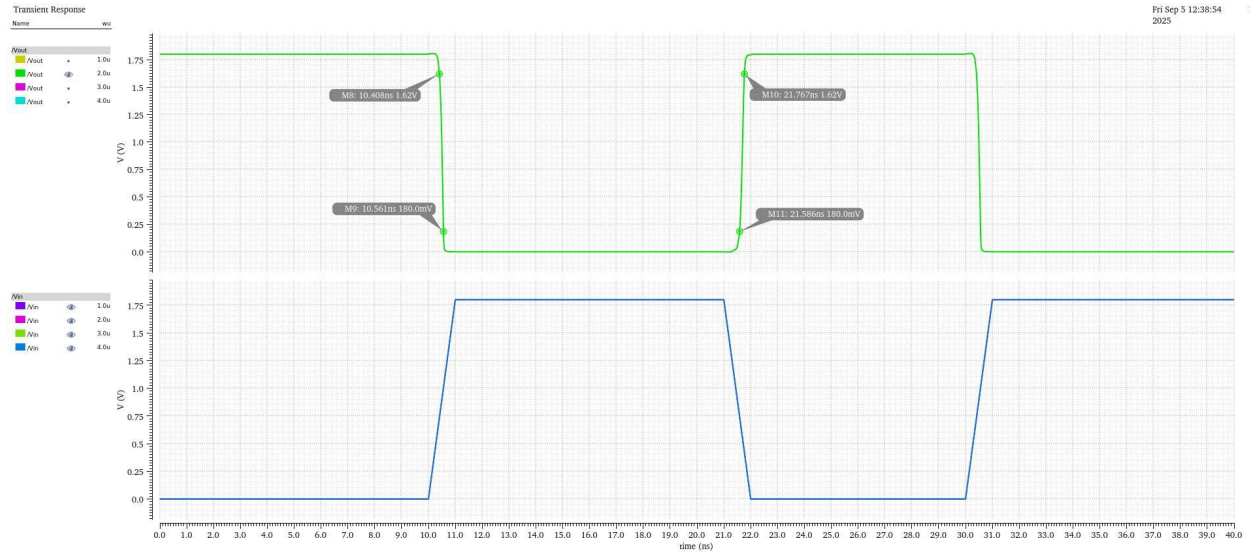
Results:

PMOS with varying width

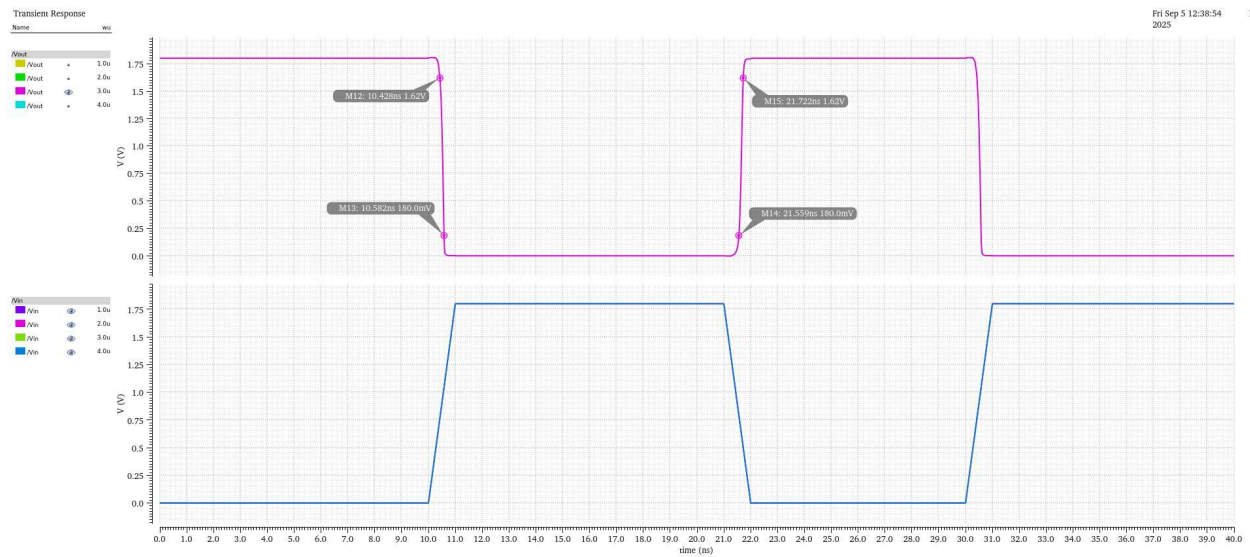
$W=1\text{ }\mu$



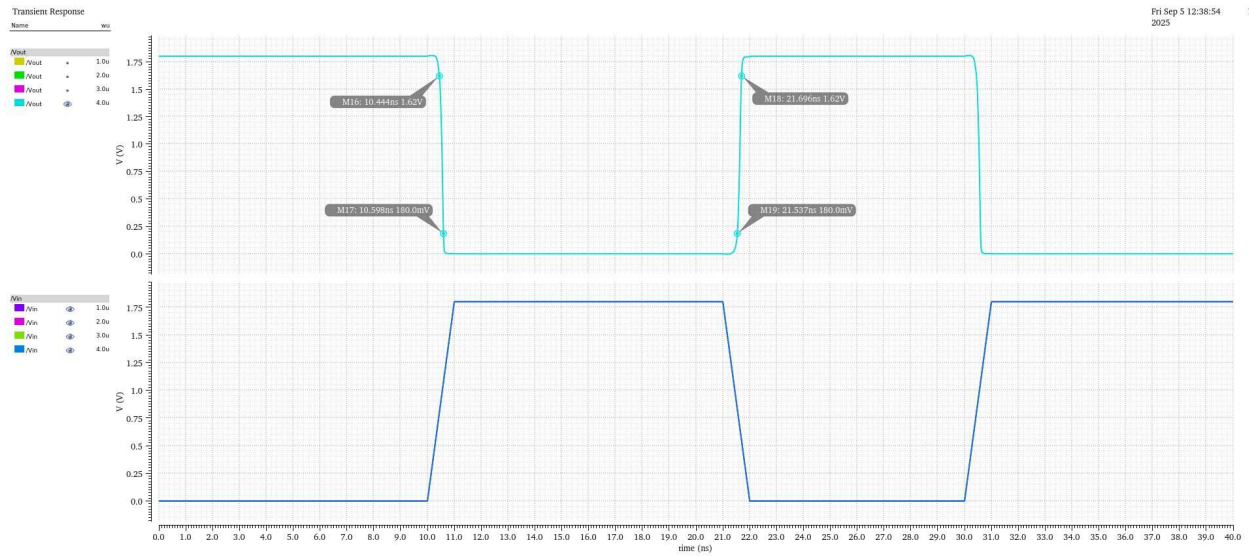
W=2u



W=3u

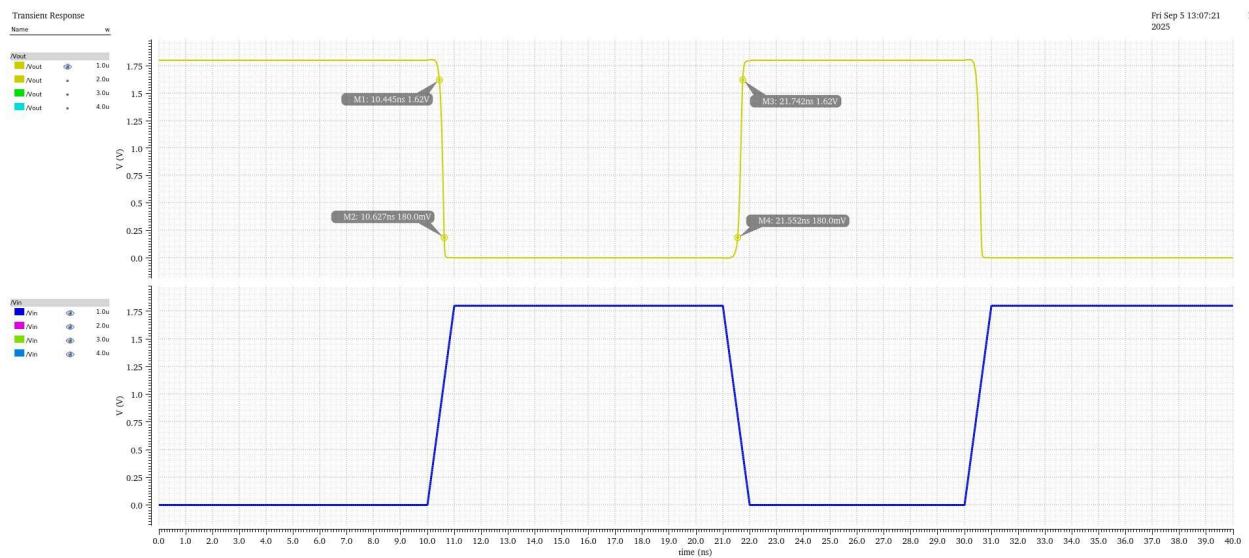


W=4u

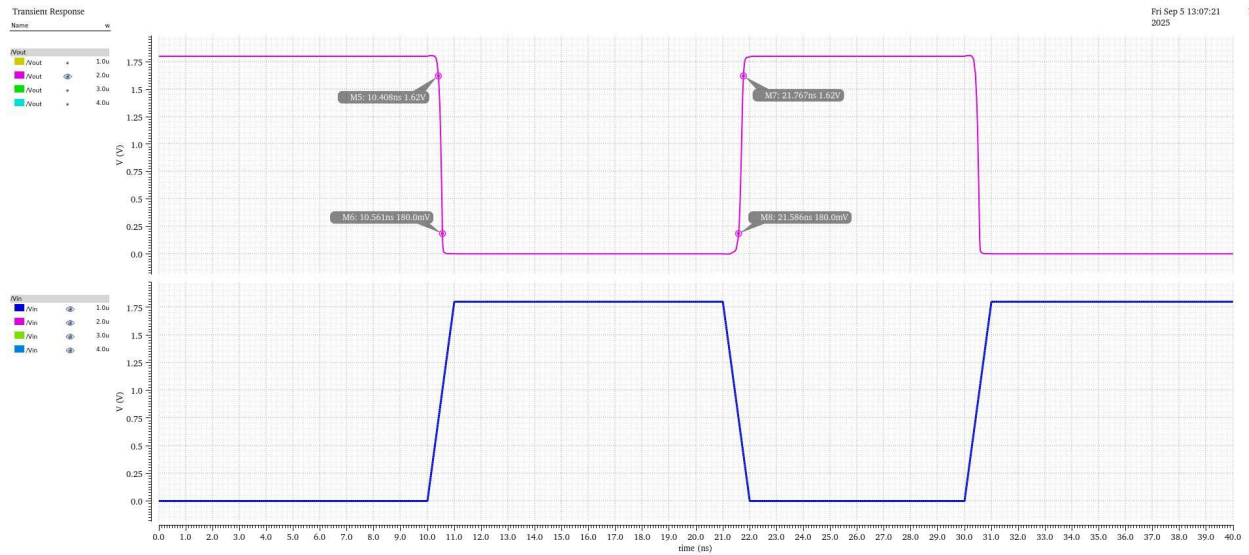


Nmos with varying width

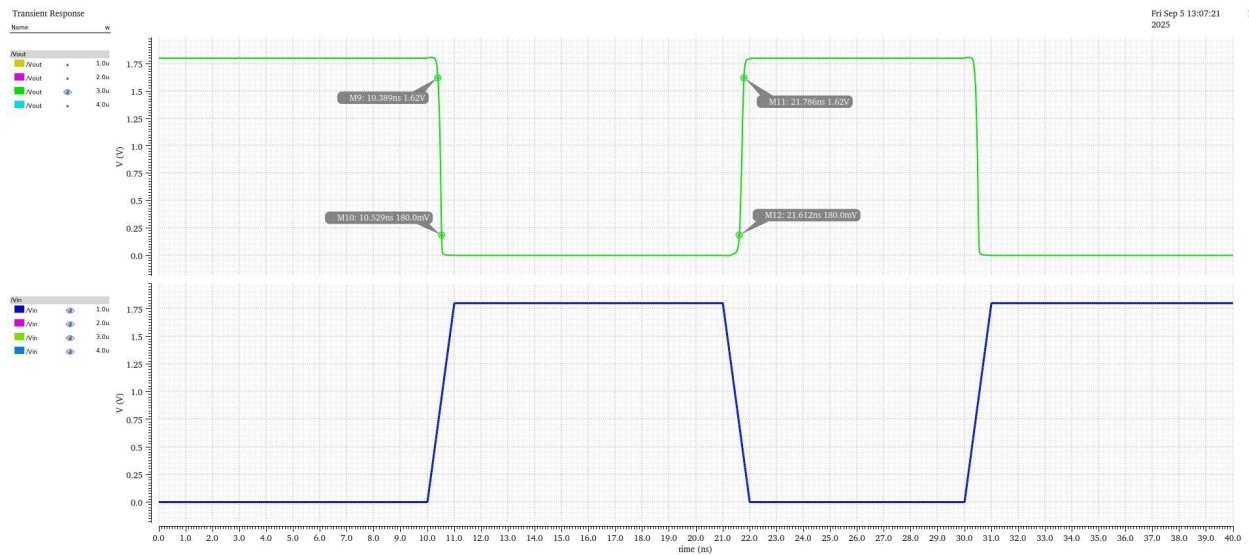
W=1u



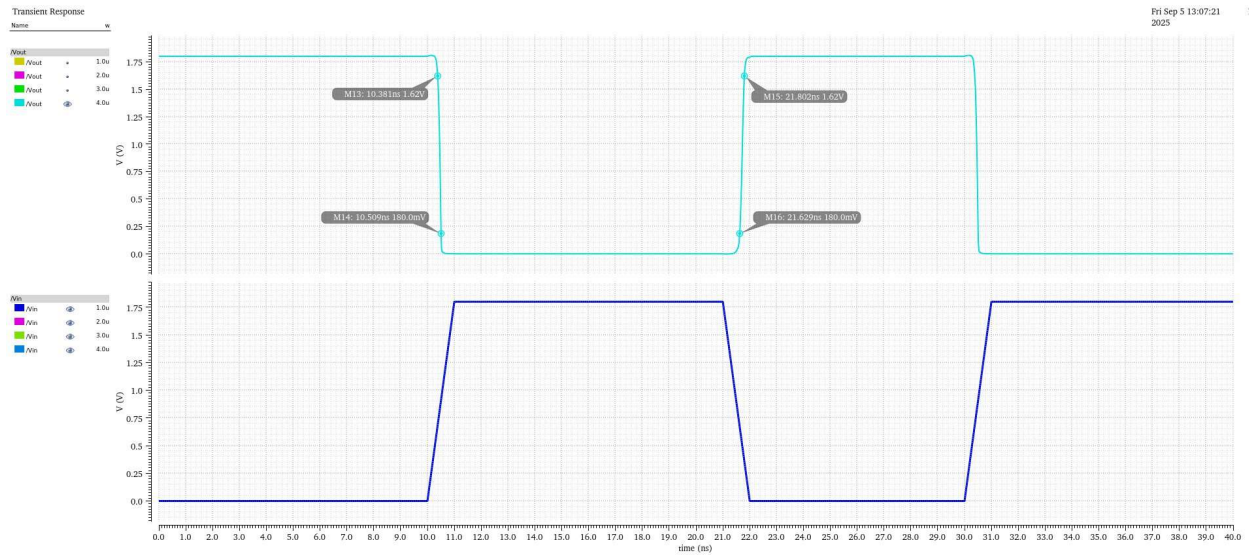
W=2u



W=3u



W=4u



Observation.

NMOS	Trise	Tfall
1u	0.19	0.182
2u	0.181	0.153
3u	0.174	0.14
4u	0.173	0.128

PMOS	Trise	Tfall
1u	0.23	0.15
2u	0.181	0.153
3u	0.163	0.154
4u	0.159	0.154

Conclusion:**PMOS Transistor:**

- With an increase in channel length, the rise time (T_{rise}) reduces, indicating faster switching of the PMOS transistor.
- The fall time (T_{fall}) shows little to no variation with changes in channel length.

NMOS Transistor:

- Like the PMOS, the rise time (T_{rise}) decreases slightly as channel length increases.
- However, the fall time (T_{fall}) drops noticeably with longer channel lengths, with a particularly sharp reduction at 4 μm

Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.